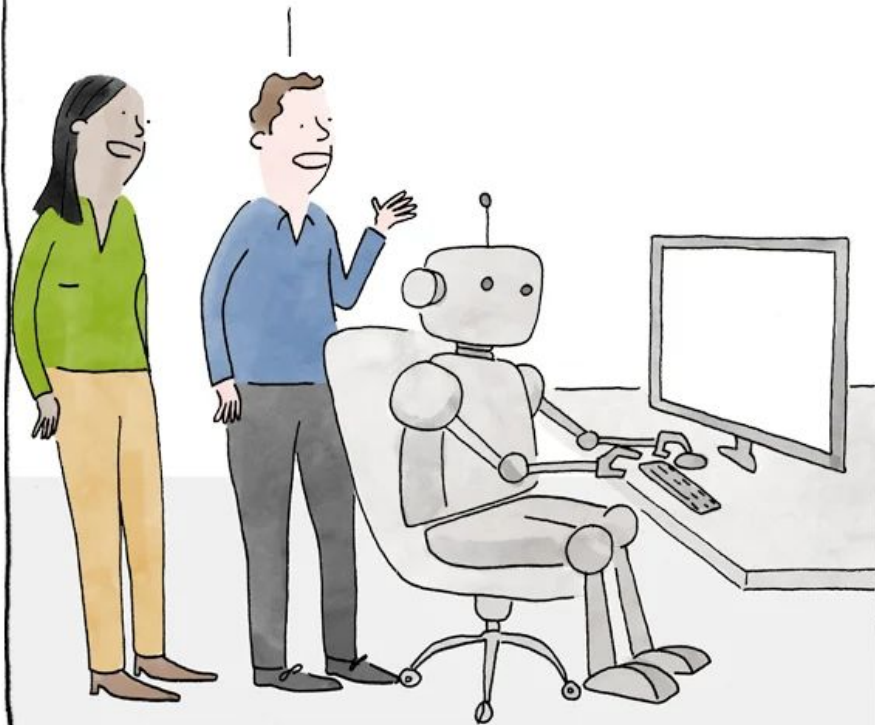
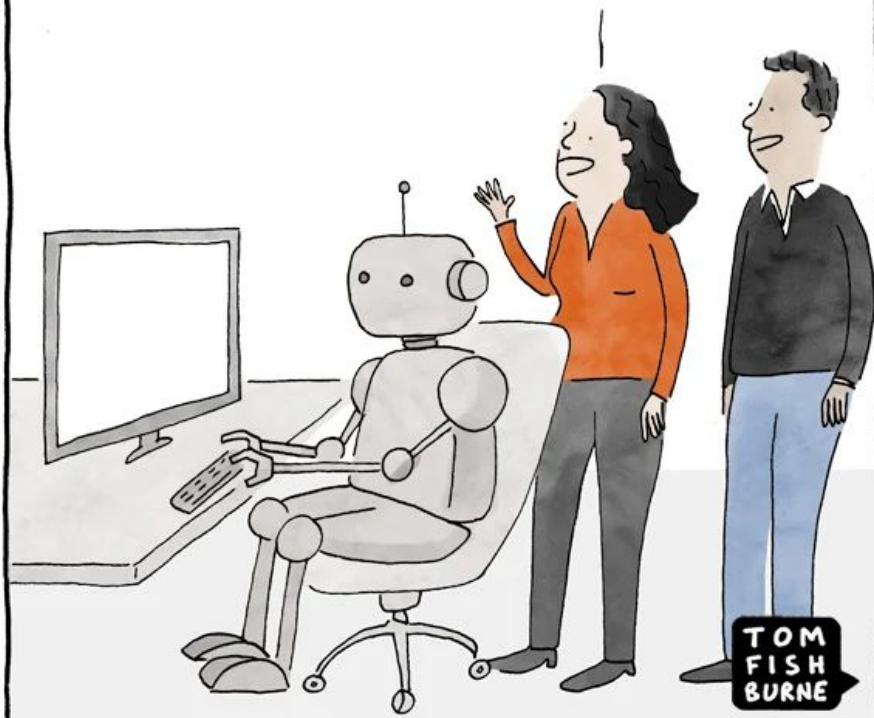


Our new Marketing AI Agent
will help us engage Customers
like never before.



Our new Customer AI Agent
will help us ignore Marketers
like never before.



TOM
FISH
BURNE



Quiz 2:

Process Analysis & Prompt Engineering

Question:

What type of step is distributing candies in class?

Question:

What type of step is distributing candies in class?

Answer:

Value Added (VA)

▶

Question:

What is the best approach for Boeing to prioritize its issues?

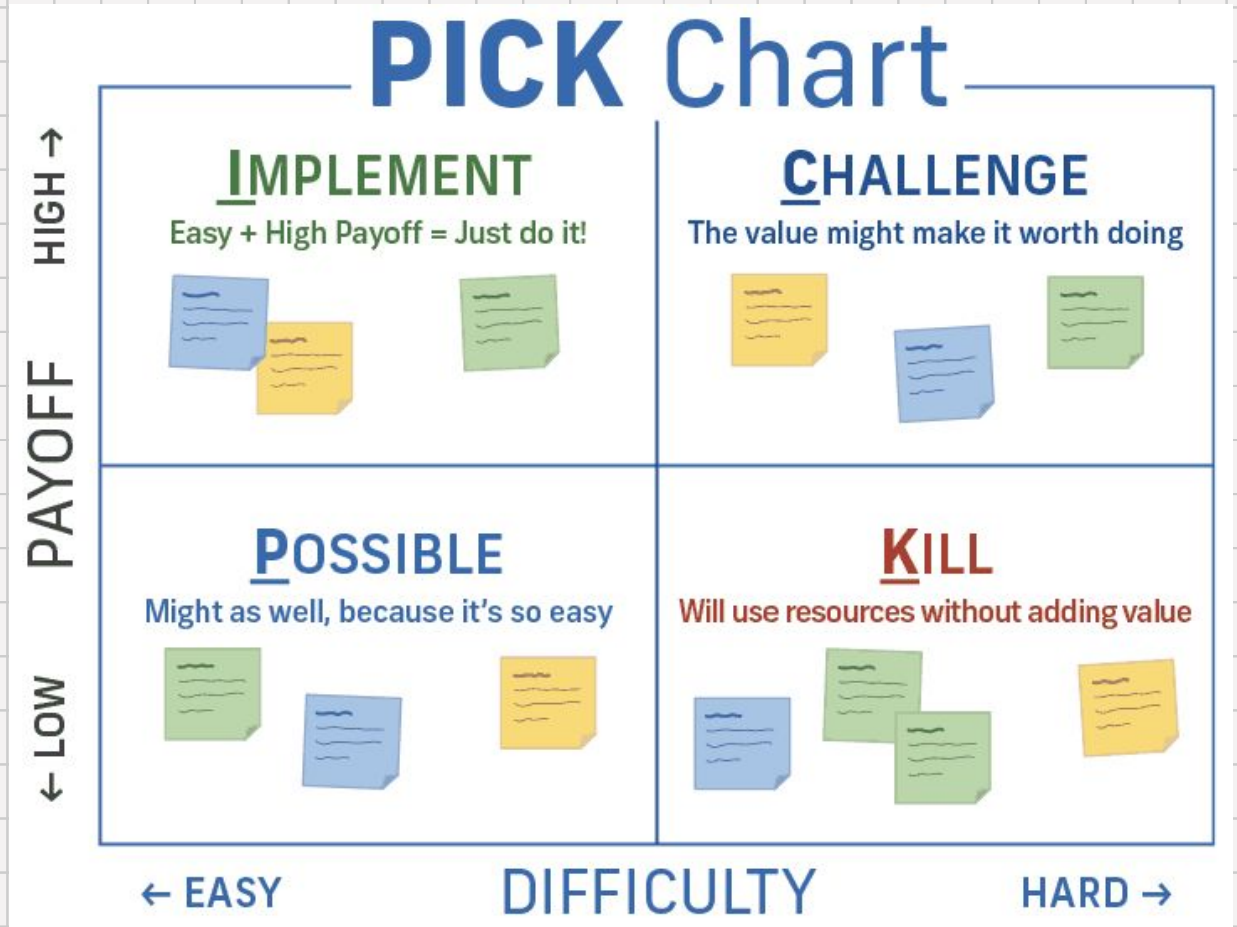
Question:

What is the best approach for Boeing to prioritize its issues?

Answer:

PICK chart

PICK Chart



Question:

Which quadrant in a PICK chart would you tackle last?

Answer:

CHALLENGE

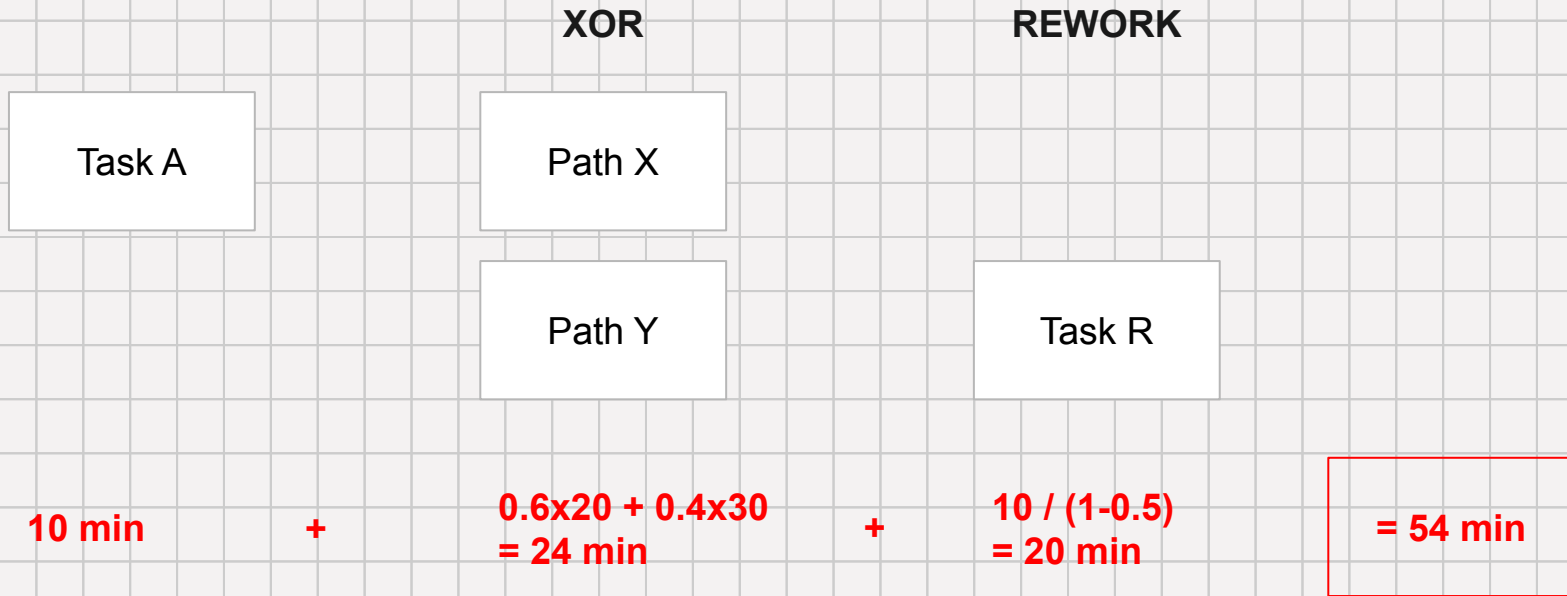


An insurance claim process uses an XOR gateway with three branches: Standard Review (p=0.5, T=10 min), Complex Review (p=0.3, T=25 min), and Escalated Review (p=0.2, T=40 min). What is the expected cycle time contribution of the XOR split alone?

$$CT = \sum_{i=1}^n p_i \times T_i$$

Branch	Probability (pi)	Time (Ti)	Pi x Ti
Standard Review	0.5	10	5.0
Complex Review	0.3	25	7.5
Escalated Review	0.2	40	8.0
		CT =	20.5 min

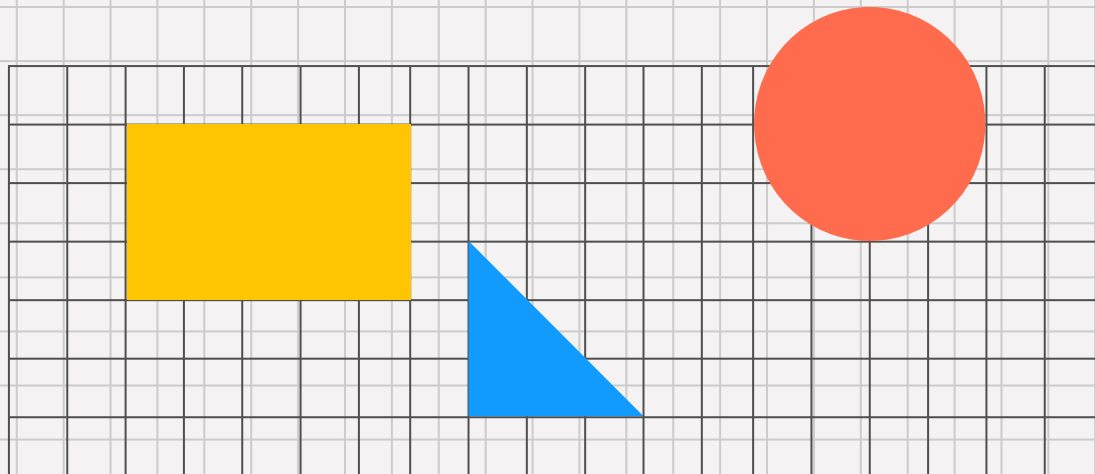
A process has Task A (10 min, sequential) → XOR: Path X (p=0.6, T=20 min) or Path Y (p=0.4, T=30 min) → Rework task R (T=10 min, r=0.5). What is the total expected cycle time?



Spring 2026

IS 365

Lecture 10: Fundamentals of Agentic AI



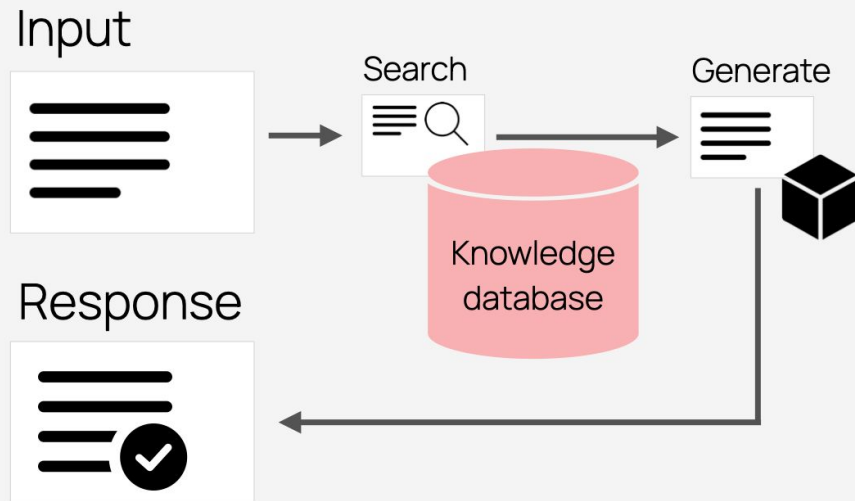


Recap

Large Language Model (LLM)

Advanced AI system capable of understanding and generating human-like text.

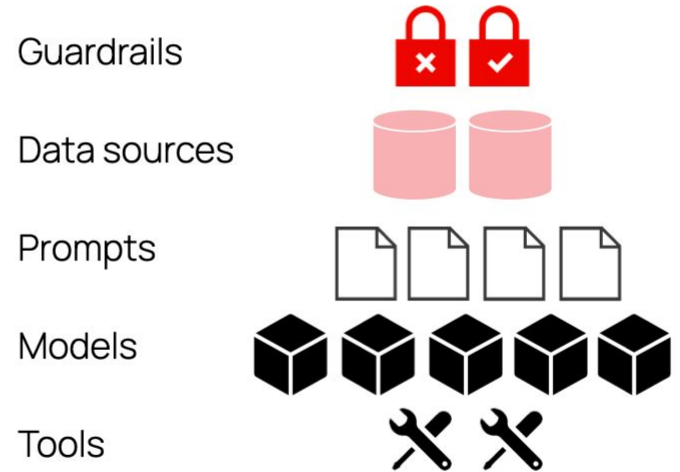
Based on “transformer model” architecture.

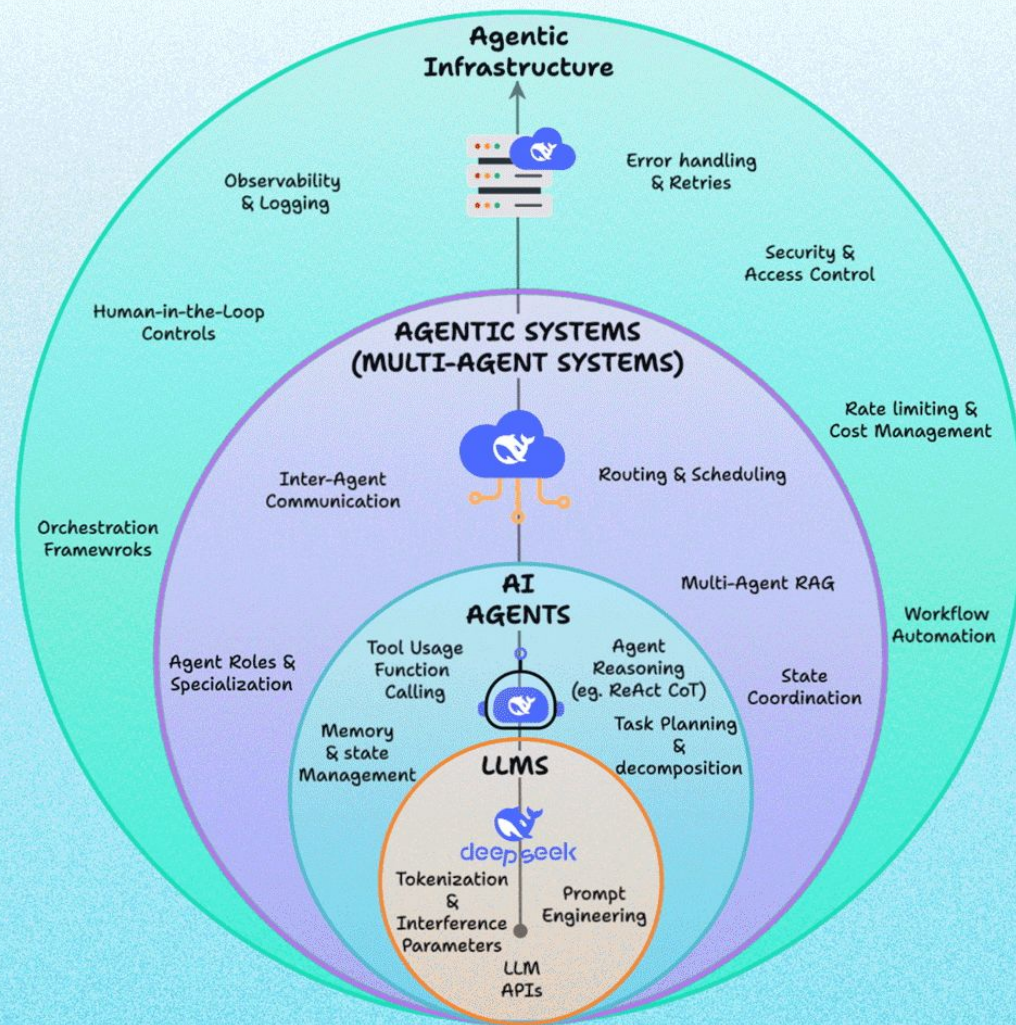


Generative AI

Produces **new content** (e.g., *text, images, and code*) in response to **user prompts** based on patterns learnt from large datasets.

Can use LLM, GAN, Diffusion, and other models.





Evolution of Agentic AI

01

Initial Conversational Agents

The first AI bots interacted with humans using natural language, but they had limited capabilities.

02

Generative AI

Generative AI emerged, improving the versatility and accessibility of AI.

03

Multimodal Advancement

AI has become multimodal, integrating multiple input and output formats.

04

Improved Reasoning

AI has improved its reasoning ability, handling complex tasks with less human intervention.

05

Greater Autonomy

AI became more autonomous and adaptable, incorporating tools such as memory and internet access.

06

Specialized Agents

Highly specialized agents were developed, demonstrating high autonomy in specific tasks.

Evolution of Agentic AI

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Reasoning & Autonomy

Advancing Reasoning in AI

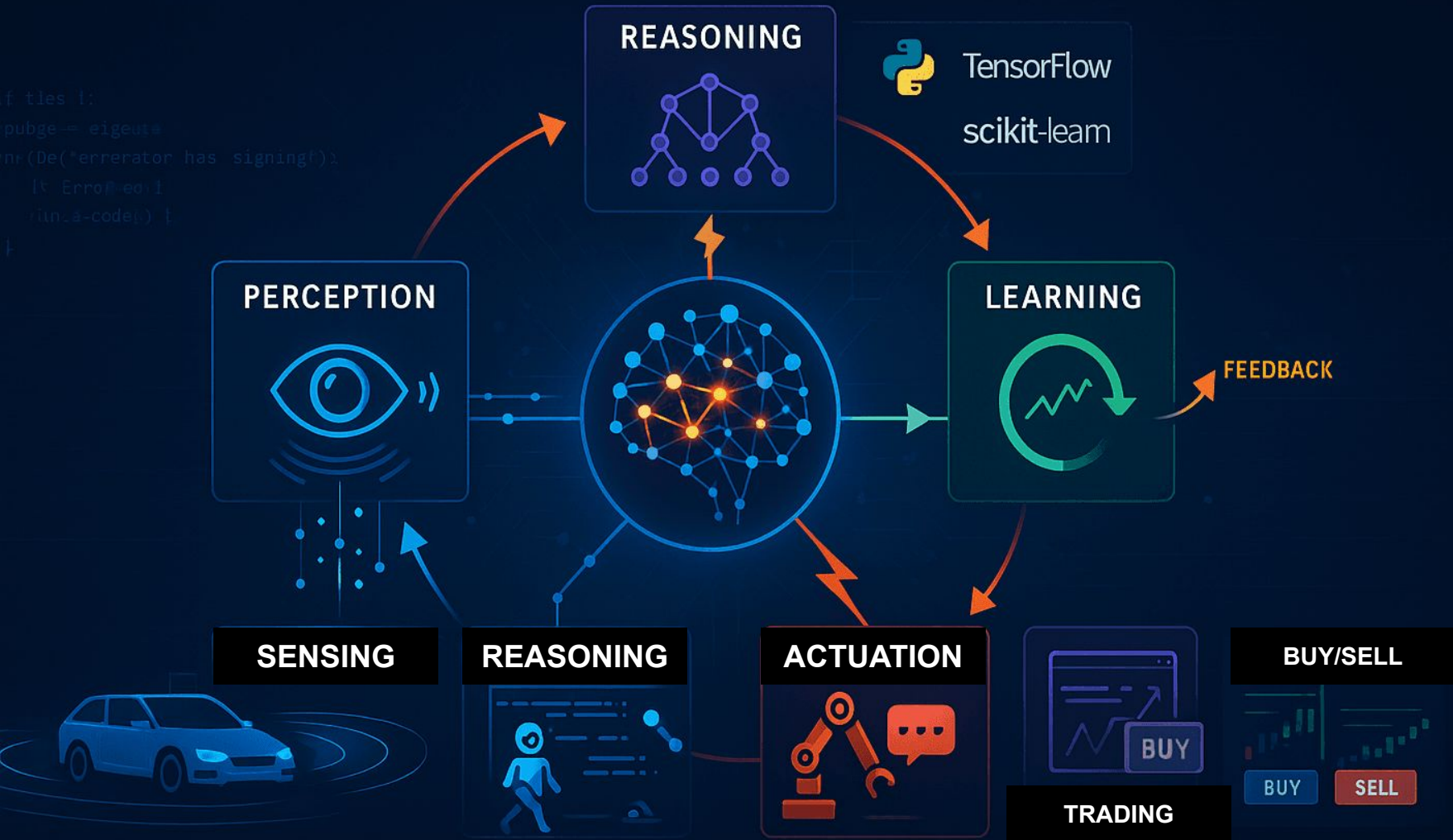
- Models like OpenAI's o1 and DeepSeek R1 improve complex reasoning
- Reduce need for human **step-by-step prompting**
- Enable **autonomous** task execution with **contextual** adaptation
- Enhance problem-solving effectiveness and consistency

Towards True Autonomy

- Generative AI now has features like memory, internet access, scheduling
- AI has transitioned from 'base agents' to autonomous systems
- Shift from user as 'producer' to 'auditor' or 'editor'
- **Examples:** OpenAI Operator, Anthropic Computer Use Agent



```
if files != []:  
    pubge = eigens  
    nr(Be('errorator has signing')):  
    { Error: eq:1  
      (un_a-code) }  
+
```



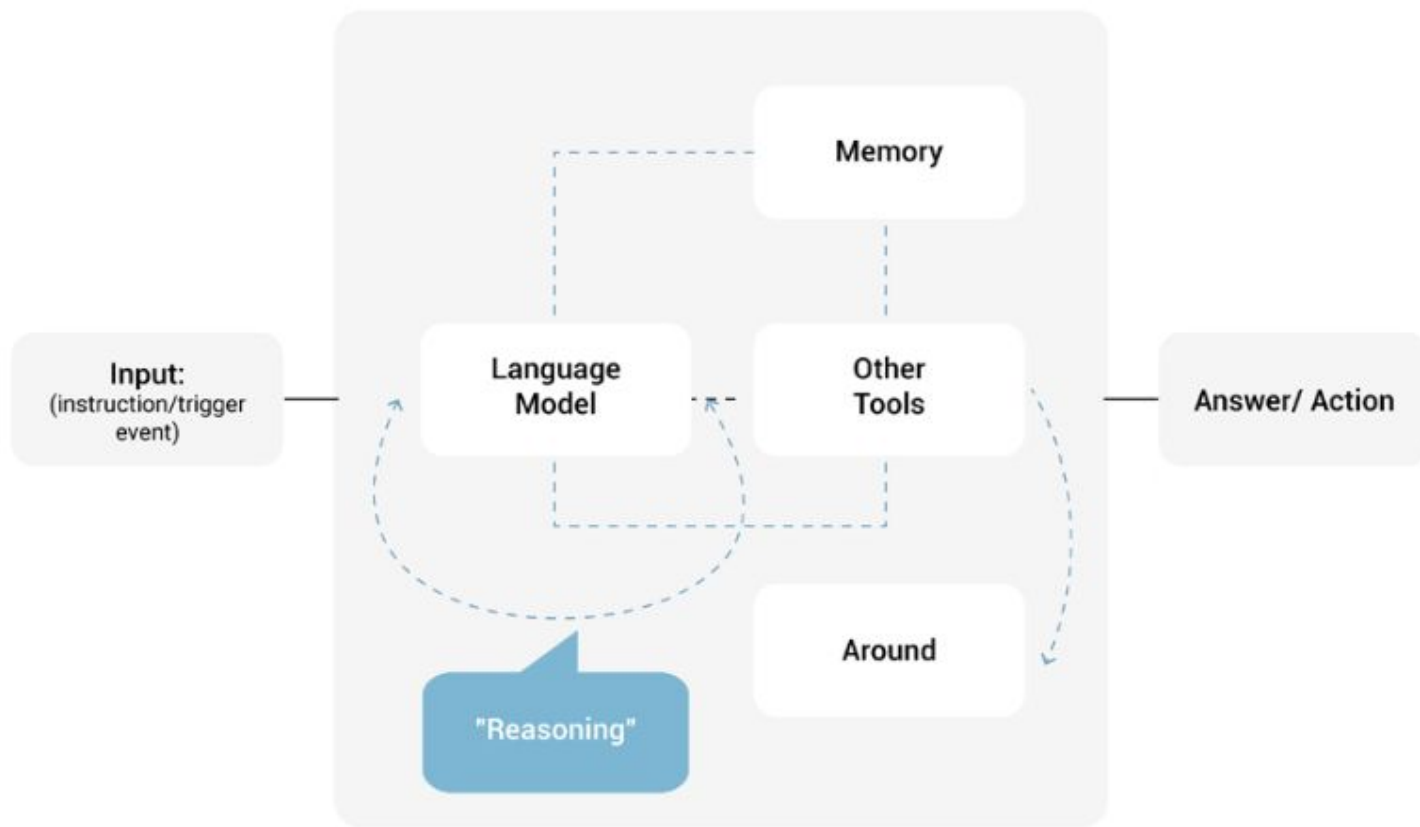
AI Agents in Practice

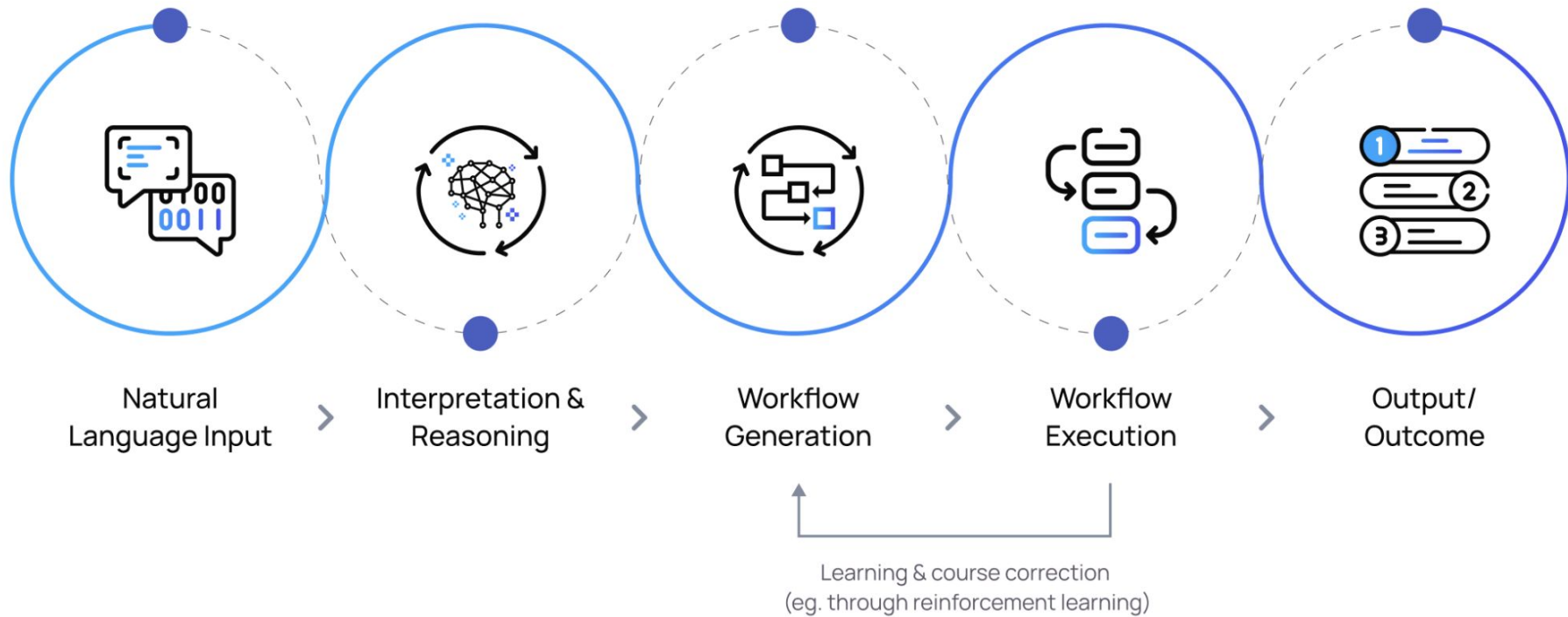
- Companies like Deloitte and Google define agents by degree of **autonomy and memory**
- Agents combine reasoning, planning, and external tool use
- Operate with high autonomy in defined contexts
- **Transforming workflows** across industries

AI Agent

What are AI Agents?

- Agents **execute complete workflows** autonomously and adaptively
- Base agents are LLMs; specific agents perform **specialized tasks**
- **Multi-agent systems** coordinate via orchestration logic
- **Example:** logistics agent managing inventory and forecasting

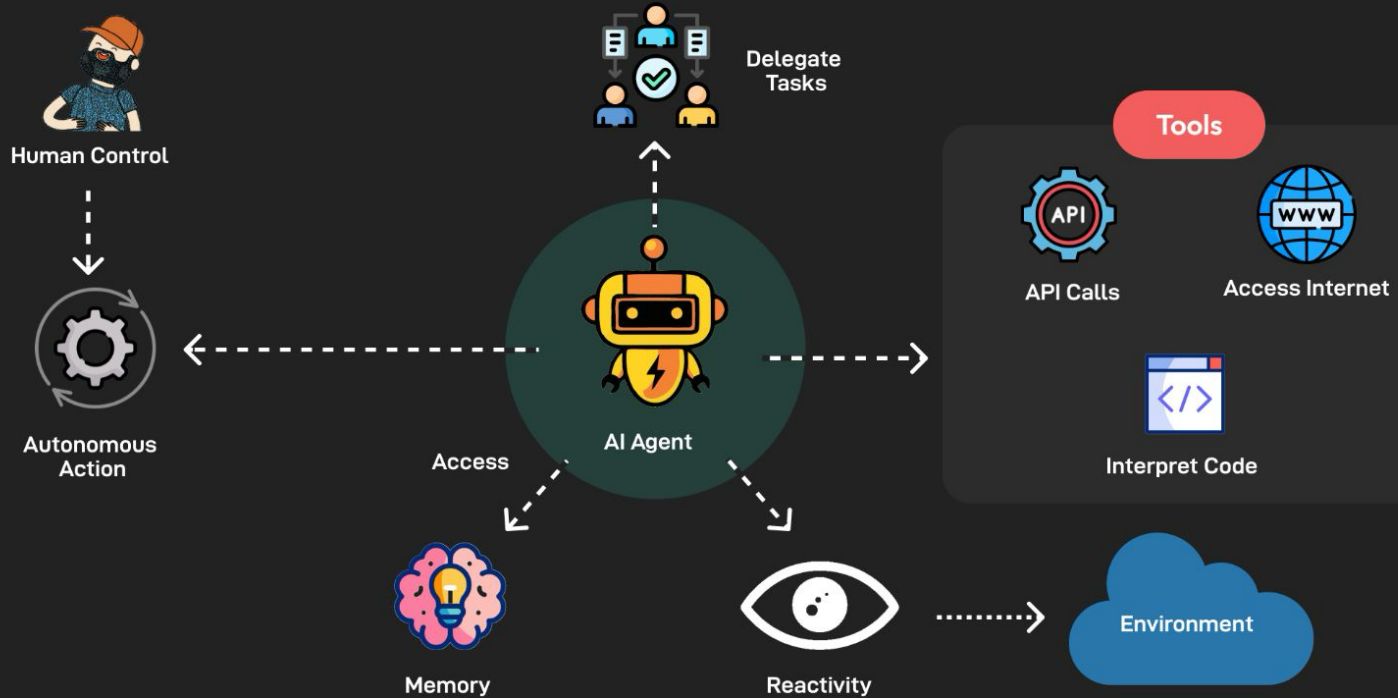




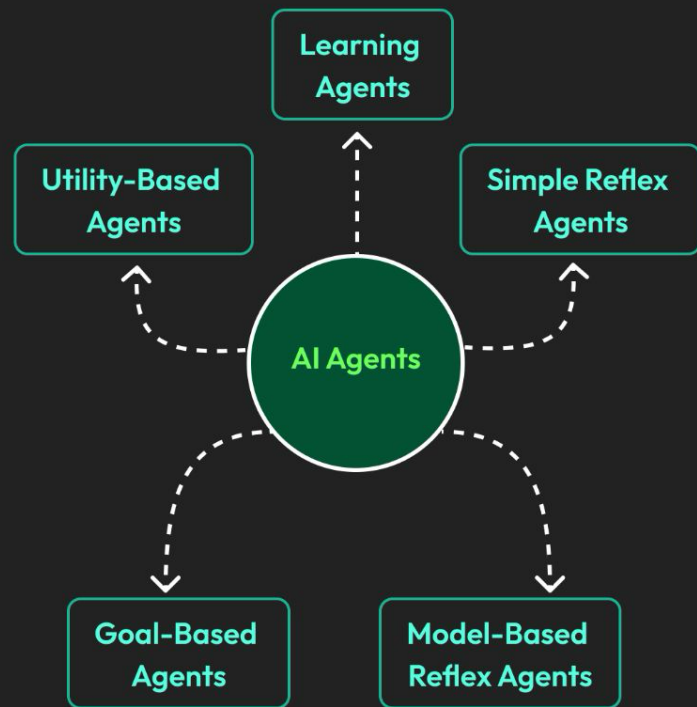
Components of AI Agents

- **Base agent (language model):** reasoning and decision-making core
- **Memory module:** retains context and long-term information
- **Tools:** APIs and external databases for expanded functionality
- **Orchestration layer:** cycles perception, reasoning, and action

How AI Agents Work?



Types of AI Agents



AI Agent System Architecture

Single Agent



Agents act as personal assistants

Multi-Agent



Agents interact with each other in collaborative ways

Human-Machine



Agents interact with humans to provide assistance

Specialized Agents

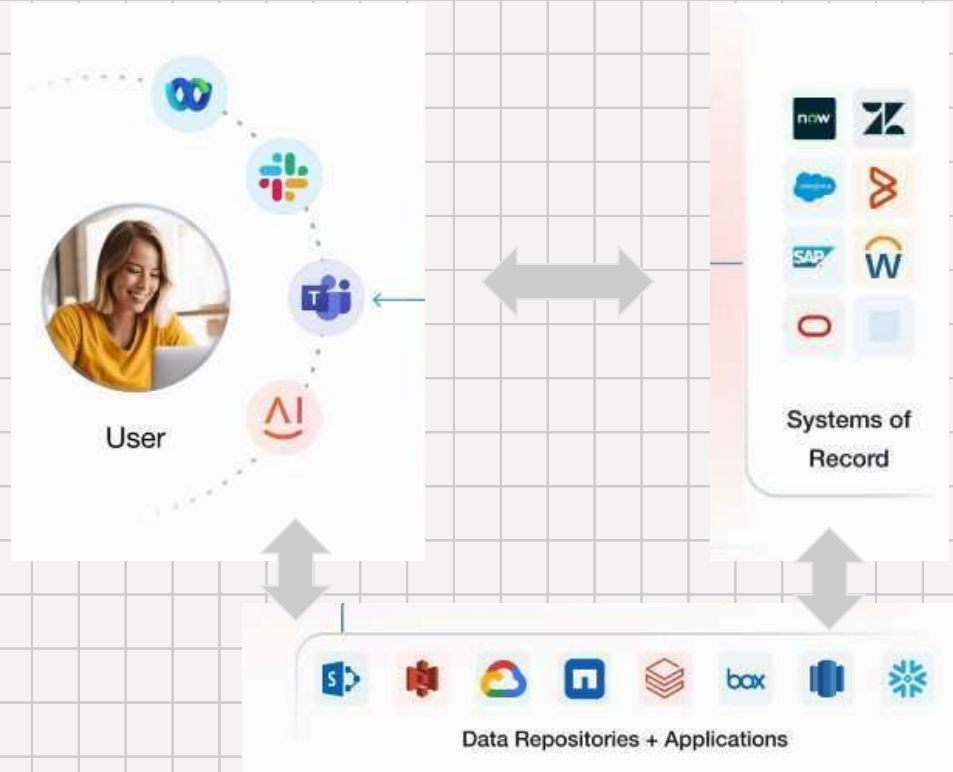
- Base agents can evolve into task-specific AI agents
- **Types:** Multitasking, Derived, and Orchestrating Agents
- Each has distinct autonomy and coordination levels
- Support complex, adaptive workflows

Examples: AI Agents

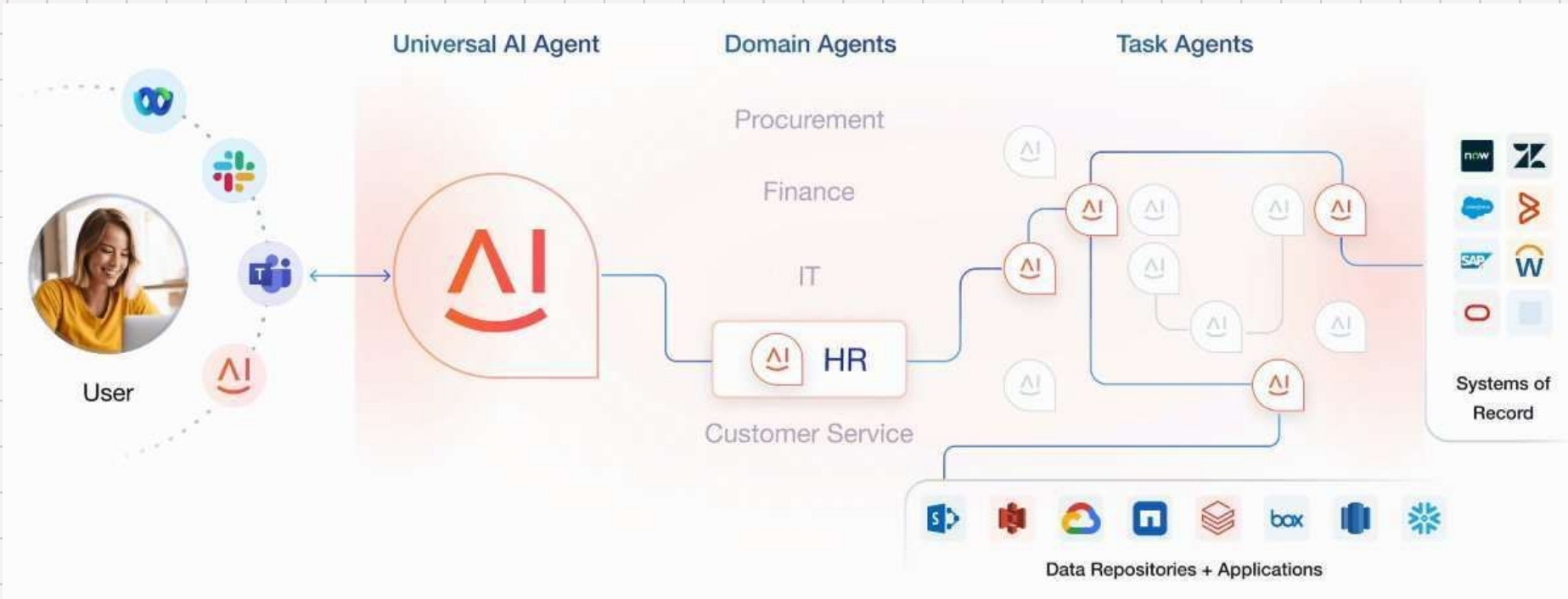
- **IT:** Detect incidents, diagnose root causes, resolve tickets
- **Finance:** Monitor transactions, flag fraud, initiate account holds / notifications
- **HR:** Onboard, provision accounts, answer policy questions
- **Customer Support:** Issue refunds / upgrades, schedule, manage help desk
- **Supply Chain:** Track inventory, reroute shipments, reorder stock based on forecasts



Current Situation



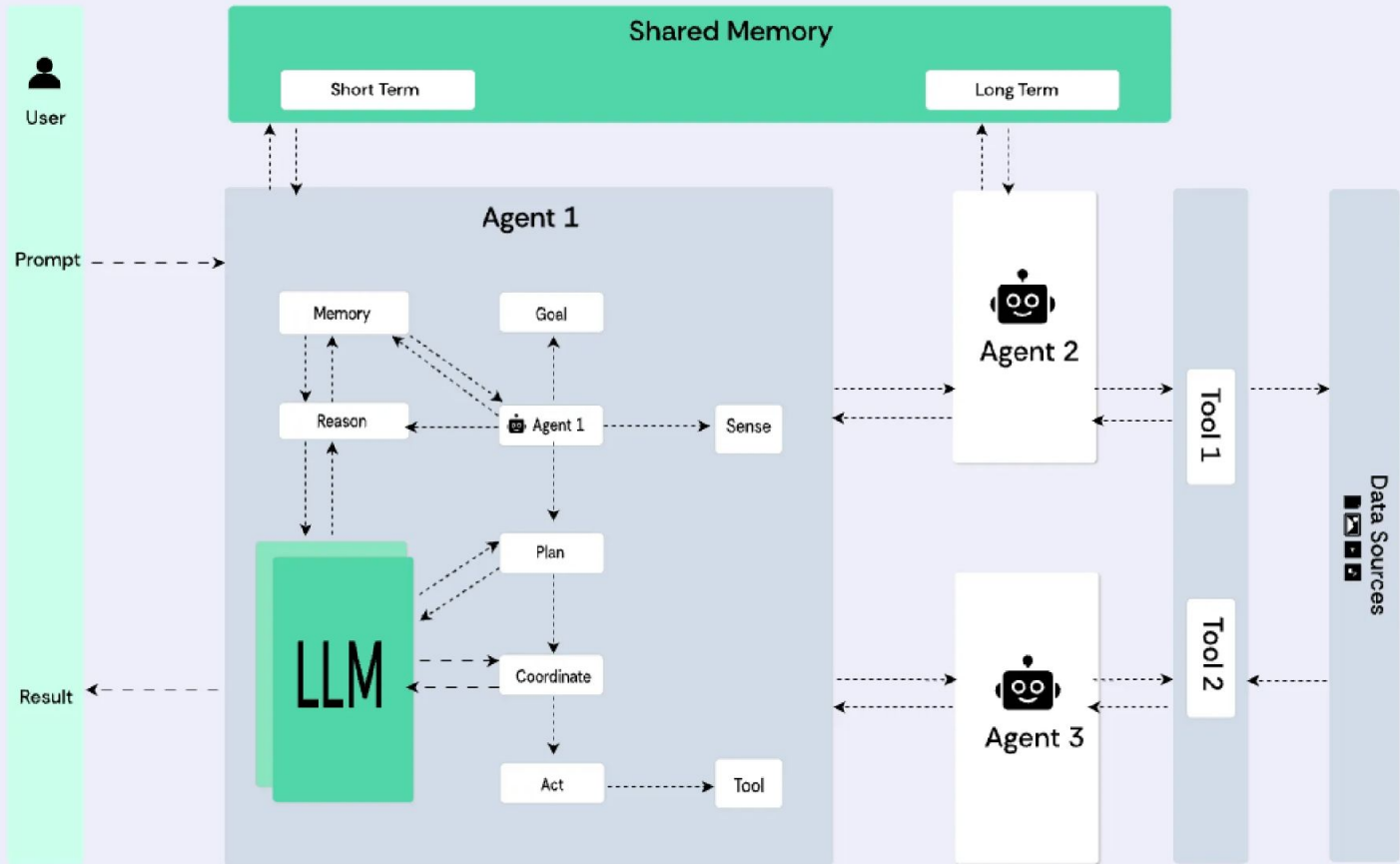
Future with AI Agents



Agentic AI

Agentic AI systems exhibit **autonomy**, **goal-orientation**, **reasoning**, and the ability to take **actions**. They plan, make decisions, and dynamically adapt to changing contexts without constant human input.

These agents can **collaborate, self-correct, and learn** from outcomes to improve over time.





AI AGENTS VS AGENTIC AI

AI AGENTS

(SPECIFIC DEFINED AI SYSTEM)

CHATBOT

CUSTOMER
SERVICE

TASK-SPECIFIC

**VIRTUAL
ASSISTANT**

ALEXA, SIRI, GROK

GENERAL,
MULTI-FUNCTIONAL

**PERSONAL
TUTOR**

DATA ANALYST

ACCESS TOOLS

MCP

SPECIFIC
USE-CASES

- CONCRETE IMPLEMENTATIONS
- SPECIFIC USE-CASES
- DEPLOYED APPLICATIONS
- TOOL / MCP FOCUSED

AGENTIC AI

(AI DESIGN PHILOSOPHY FOR APPLICATIONS)

AUTONOMY

SELF-DIRECTED ACTION

EMBEDDED

INTEGRATED

APPROACH

GOAL ORIENTED

PURPOSE-DRIVEN

AUTONOMOUS
RESEARCH ASSISTANT

ADAPTABILITY

OBSERVE, LEARN,
ADJUST

REASONING

REASON, PLAN &
ACT

SELF-MANAGEMENT
SYSTEM

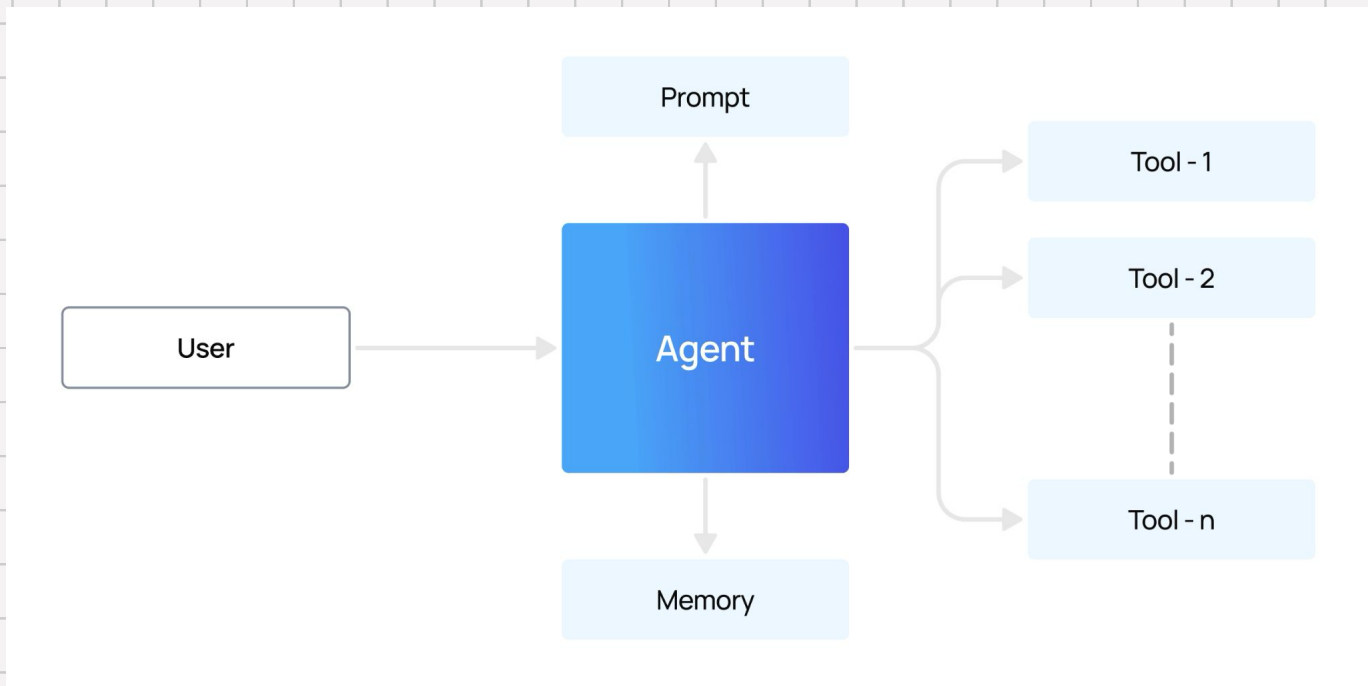
- INDEPENDENT DECISION-MAKING
- DYNAMIC GOAL PURSUIT
- ENVIRONMENT AWARENESS FOR TRIGGERS
- EMERGENT BEHAVIOUR
- APPROACH-FOCUSED DESIGN

Agentic AI Architectures

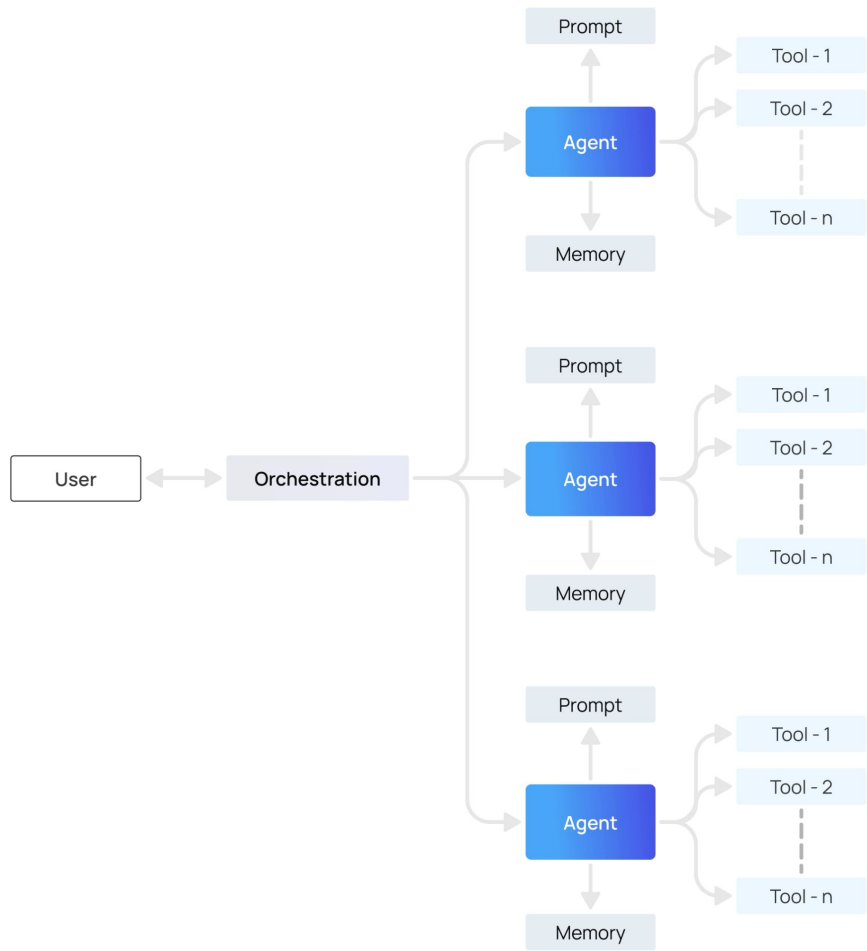
- **Single-agent systems:** isolated task performance
- **Multi-agent systems:** collaboration between agents
- **Hierarchical models:** supervisor agent coordinates tasks
- **Sequential systems:** chained, dependent task execution

Feature	Agentic AI	Generative AI	Traditional AI
Primary Function	Goal-oriented action & decision-making	Content generation (text, code, images, etc.)	Focused on automating repetitive tasks
Autonomy	High – Operates with minimal human oversight	Variable – May require user prompts or guidance	Low – Relies on specific algorithms and set rules
Learning	Reinforced Learning – Improves through experience	Data-driven learning – Learns from existing data	Relies on predefined rules and human intervention

One AI Agent



Agentic AI System



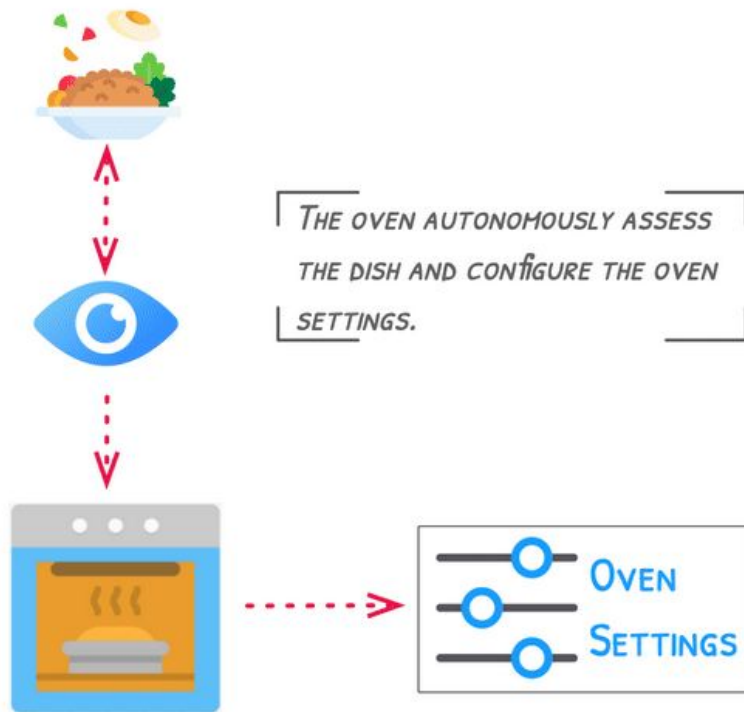
Use Case: Project Management

- Multi-agent system for managing complex projects
- **Agents** assigned for data analysis, planning, and reporting
- **Orchestrator** ensures coordination and goal tracking
- Improves efficiency and consistency in project outcomes



SMART KITCHEN SYSTEM

AI AGENT



(SINGLE TASK)

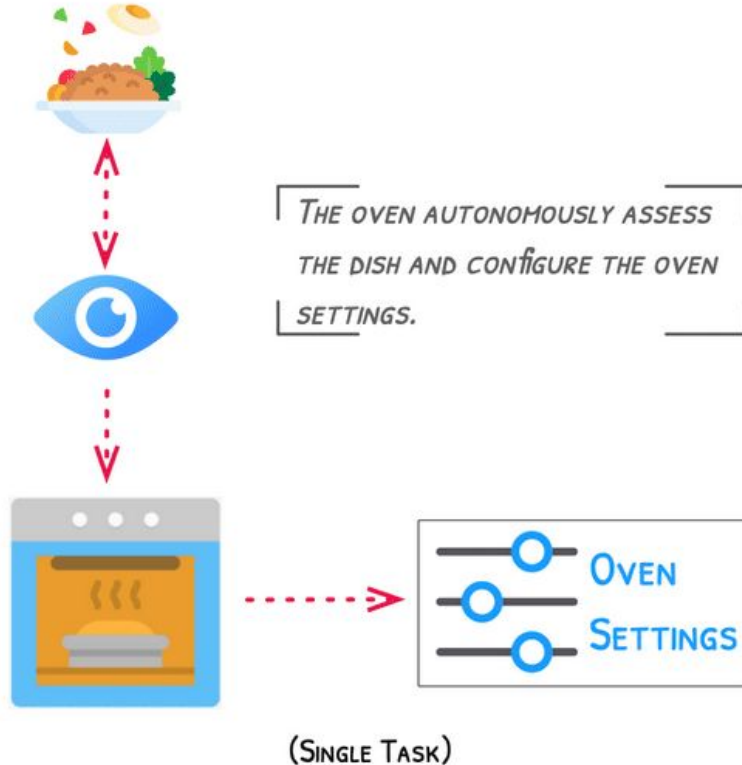
AGENTIC AI



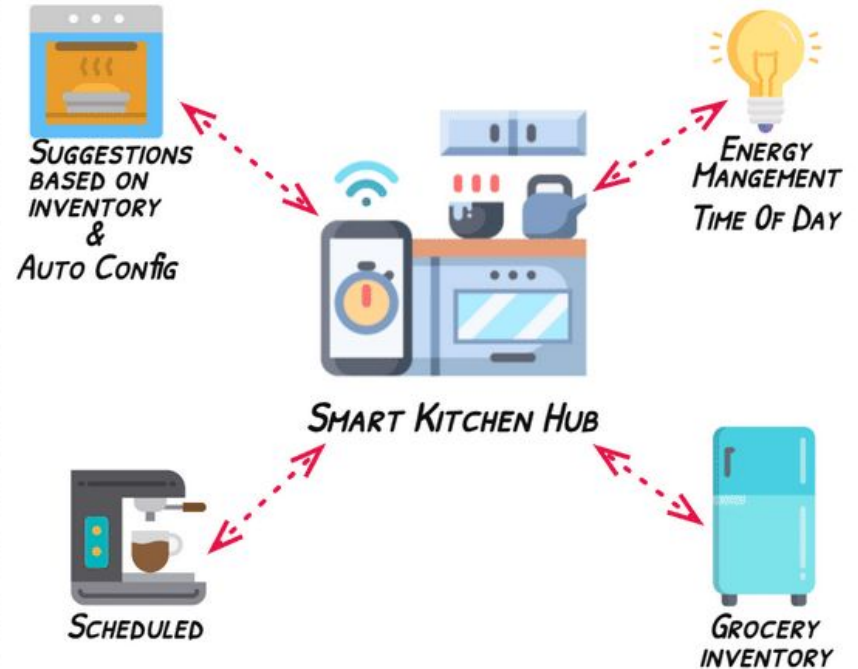
(MULTI-AGENT, COLLABORATIVE AGENTIC AI SYSTEM)

SMART KITCHEN SYSTEM

AI AGENT



AGENTIC AI



(MULTI-AGENT, COLLABORATIVE AGENTIC AI SYSTEM)

General Agentic AI Products

Company	Key Agentic Product(s)	Agentic Features	Pricing Model
OpenAI	GPT-4.1 / GPT-4o, Operators API	Planning, tool use, memory	Pay-per-token
Microsoft	Autogen, Copilot Studio, Azure AI Agents	Multi-agent orchestration	Subscription + token use
Amazon	AWS Bedrock Agents, Amazon Q	Enterprise API/action agents	Usage-based
Google	Gemini 1.5 Pro, Vertex AI Agents	Context reasoning, Workspace integration	Subscription + usage
Meta	Meta AI Assistant, Llama 3 Agents	Open-source, multimodal agents	Free / open-source
Salesforce	Einstein 1 Agents, Copilot Studio	Business process orchestration	CRM-based pricing
Zapier	Zapier Agents	Workflow automation across 8k apps	\$0–\$50/month
Agentec	Agentec OS for Agents	Build/deploy agents with APIs	\$0–\$49/month + usage
Spark Tools	Agentic AI Framework	No-code automation agents	\$49–\$149/month
Mindset AI	Embedded Agent SDK	Custom embedded agents	~US\$3K+/month

Business Process Automation Tools

Tool	Company	Best Use Case in Business	Coding Skill Needed	Free Tier
Taskade	Taskade Inc.	Workflow automation, project management, team AI agents	🟢 None	✅ Yes
n8n	n8n GmbH	Business process automation, CRM/ERP connectors	🟡 Low-code	✅ Yes (self-hosted)
Zapier	Zapier Inc.	Connecting cloud business apps (CRM, HR, marketing)	🟢 None	✅ Yes (100 tasks/month)
Langflow	Langflow (Open Source)	Building intelligent business chatbots and agent pipelines	🟡 Low-code	✅ Yes
CrewAI	Open Source / community-driven	Simulating business teams of AI agents (sales, support, strategy)	🟢 Medium (Python)	✅ Yes